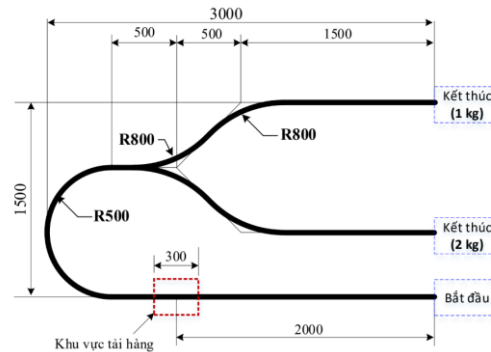
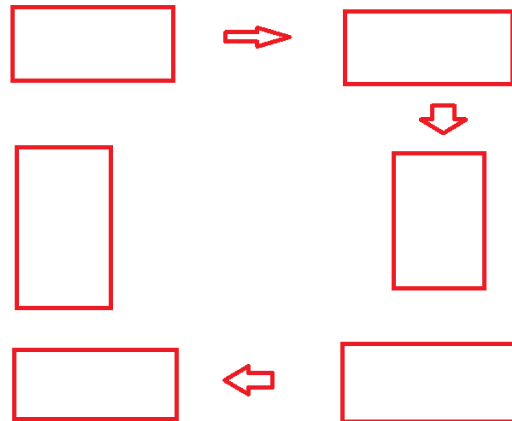


Exercise 1

1. Working Space for mobile robot is in following Figure (2 marks)



- Feature needing to watch on robot has diameter 1mm. Chose appropriate Basler Camera and working distance. Assume that the camera is setup above and perpendicular to the working space. Remember to check the result on the Basler website.
- Draw a rectangle on the window and make it move as shown



2. A picture has the matrix as following with STT is the index of the group (2 marks)

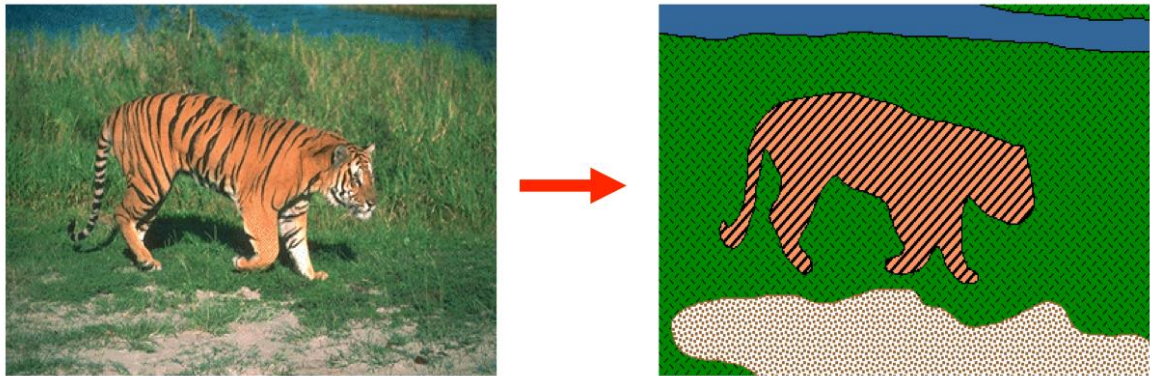
STTx10	20	30	STTx10
STTx10	STTx10	100	110
STTx10	STTx10	STTx10	150
STTx10	230	240	250

- Equalize Histogram this picture in range 0-255, using 256 bin
- Program using opencv lib to equalize histogram this picture and display histogram before and after equalize
- Threshold this picture using Otsu method with random initial guessing, Check again using opencv lib.

3. A picture has the matrix as following (2 marks)

20	22	22	35	39	126
198	20	98	20	56	97
18	22	203	99	102	101
32	201	198	98	100	201
197	199	203	101	22	198
78	200	46	250	89	129

- a) Segment this matrix into 3 groups manually and eliminate the noise.
- b) Program using opencv lib to segment this matrix and eliminate the noise
- c) Program using opencv lib to segment this picture into 3 groups as shown



4. Different pictures are shown in following figures (2 marks)



a) Program to store the number of white pixel on the circle with different Radius (1,5,10,15,20,25,30,35,40) and center of the circle is at the center of the picture to the TXT file (remember to scale the pictures to the same size)

b) Compare to identify the chessman

5. Extract color of mangos and determine theirs mature (2 marks)

